10.1 How our food is produced

This chapter will explain:

★ the different types of farming
★ the systems used in farming.

Farming types

Arable, pastoral and mixed farming

Arable farms (Figure 10.1) cultivate crops and are not involved with livestock. An arable farm may concentrate on one crop (monoculture) such as wheat, or may grow a range of different crops. The crop grown on an arable farm may change over time. For example, if the market price of potatoes increases, more farmers will be attracted to grow this crop.

▲ **Figure 10.1** Arable farming in the Nile valley, with the pyramids in the background

Pastoral farming (Figure 10.2) involves keeping livestock such as dairy cattle, sheep, goats and pigs.

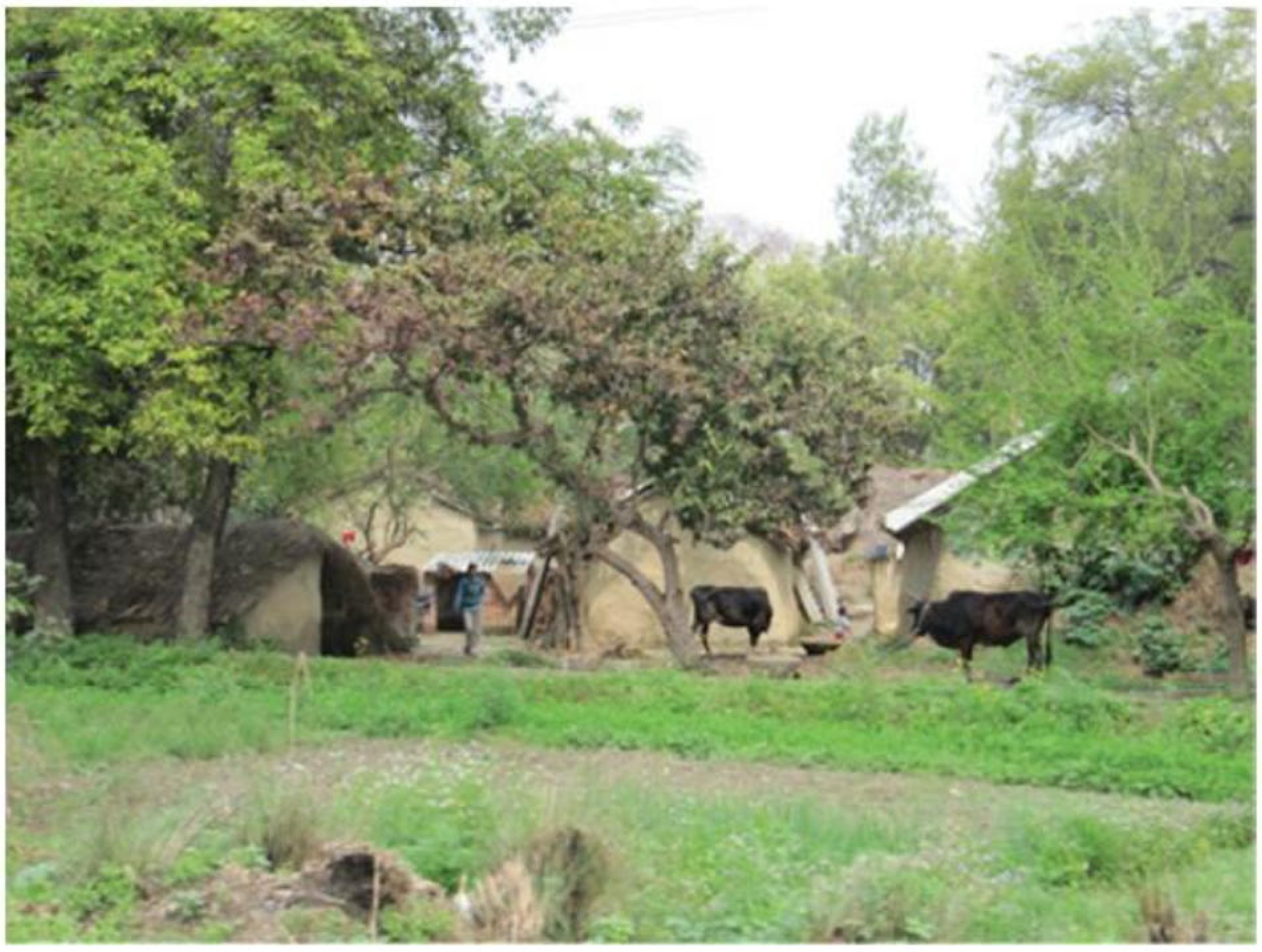
Mixed farming involves cultivating crops and keeping livestock together on a farm. Usually on a mixed farm at least part of the crop production will be used to feed the livestock.

▲ **Figure 10.2** Pastoral farming – goats feeding from a bowl (because the ground is frozen) in Central Asia

Subsistence and commercial farming

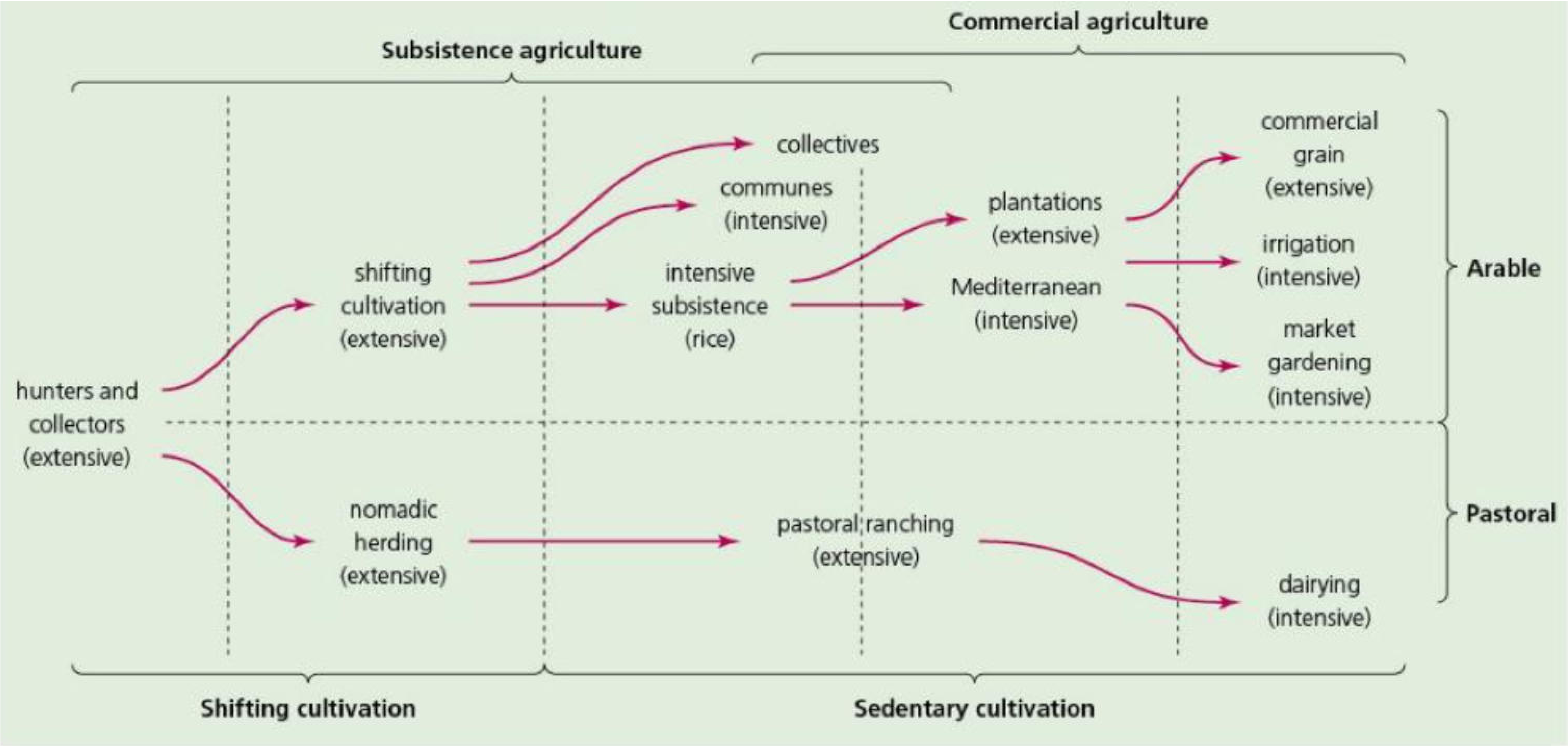
Subsistence farming is the most basic form of agriculture, in which the produce is consumed entirely or mainly by the family who work the land or tend the livestock. Figure 10.3 shows a smallholding in northern India with less than half a hectare of land. The family have two cows and grow crops for themselves and to feed their cows. The manure from the cows is dried and used as a fuel for cooking (and for heating in the cold months). If a small surplus is produced by subsistence farming, it may be sold or traded. Examples of subsistence farming are shifting cultivation and nomadic pastoralism. Subsistence farming is generally small scale and labour intensive with little or no technological input.

In contrast, the objective of **commercial farming** is to sell everything that the farm produces. The aim is to maximise yields in order to achieve the highest profits possible. Commercial farming can vary from small scale (Figure 10.4) to very large scale. The very largest farms are often owned by TNCs. Figure 10.5 shows how farming types can change with levels of development.



▲ **Figure 10.3** A smallholding in northern India

▲ **Figure 10.4** A small rice farm in Sri Lanka



▲ **Figure 10.5** Farming types and levels of development

Extensive and intensive farming

Extensive farming is where a relatively small amount of agricultural produce is obtained per hectare of land, meaning such farms tend to cover large areas of land. Inputs per unit of land are low. Extensive farming can be arable or pastoral in nature. Examples of extensive farming are sheep farming in Australia and wheat cultivation on the Canadian Prairies (Figure 10.6).

In contrast, **intensive farming** is characterised by high inputs per unit of land to achieve high yields per hectare. Examples of intensive farming include market gardening, viticulture (wine production), dairy farming and horticulture. Intensive farms tend to be relatively small in terms of land area. Countries well-known for intensive farming include the Netherlands and Denmark.

▲ **Figure 10.6** Extensive farming in the Canadian Prairies (the province of Manitoba), where most farms are over 1000 hectares in size

Organic farming

Organic farming does not use manufactured chemicals, so production does not involve the use of chemical fertilisers, pesticides, insecticides or herbicides. Instead, animal and green manures are used, along with mineral fertilisers such as fish and bone meal. Organic farming therefore requires a higher input of labour than mainstream farming. Weeding is a major task in this type of farming. Organic farming is less likely to result in soil erosion and is less harmful to the environment in general. For example, there will be no nitrate runoff into streams and so much less harm to wildlife.

Organic farming tends not to produce the ‘perfect’ potato, tomato or carrot. However, because of the increasing popularity of organic produce, it commands a substantially higher price than mainstream farm produce.

Aeroponics, hydroponics and aquaponics

Advanced farming systems such as **aeroponics** and **hydroponics** allow crops to be grown in places where traditional farming cannot be practised. Neither of these relatively new systems uses soil for crop production.

» In aeroponics, plants are never placed into water and crops grow suspended in the air. Nutrients are supplied from a mist that is sprayed on to their roots. The entire process is enclosed, so the nutrient mist

is able to remain around the plants for longer. This helps plants grow more quickly than on a traditional outdoor farm.
- In hydroponics, plants may be suspended in nutrient water all the time or fed by an intermittent flow of water. Because the plants' roots are submerged, they do not receive as much oxygen as in aeroponics systems, leading to a generally smaller plant and crop yield.

Aquaponics is a system that combines fish farming (aquaculture) with hydroponics. Fish waste is naturally converted into nutrients for plants, while the plants filter the water for the fish.

Aeroponics and hydroponics are frequently practised in vertical farming systems (Figure 10.7), which can be established in relatively small areas of land. Locations are often within urban areas. The initial investment for these types of farming can be substantial, but the potential is growing for vertical farming to become a profitable venture practised on a large scale.

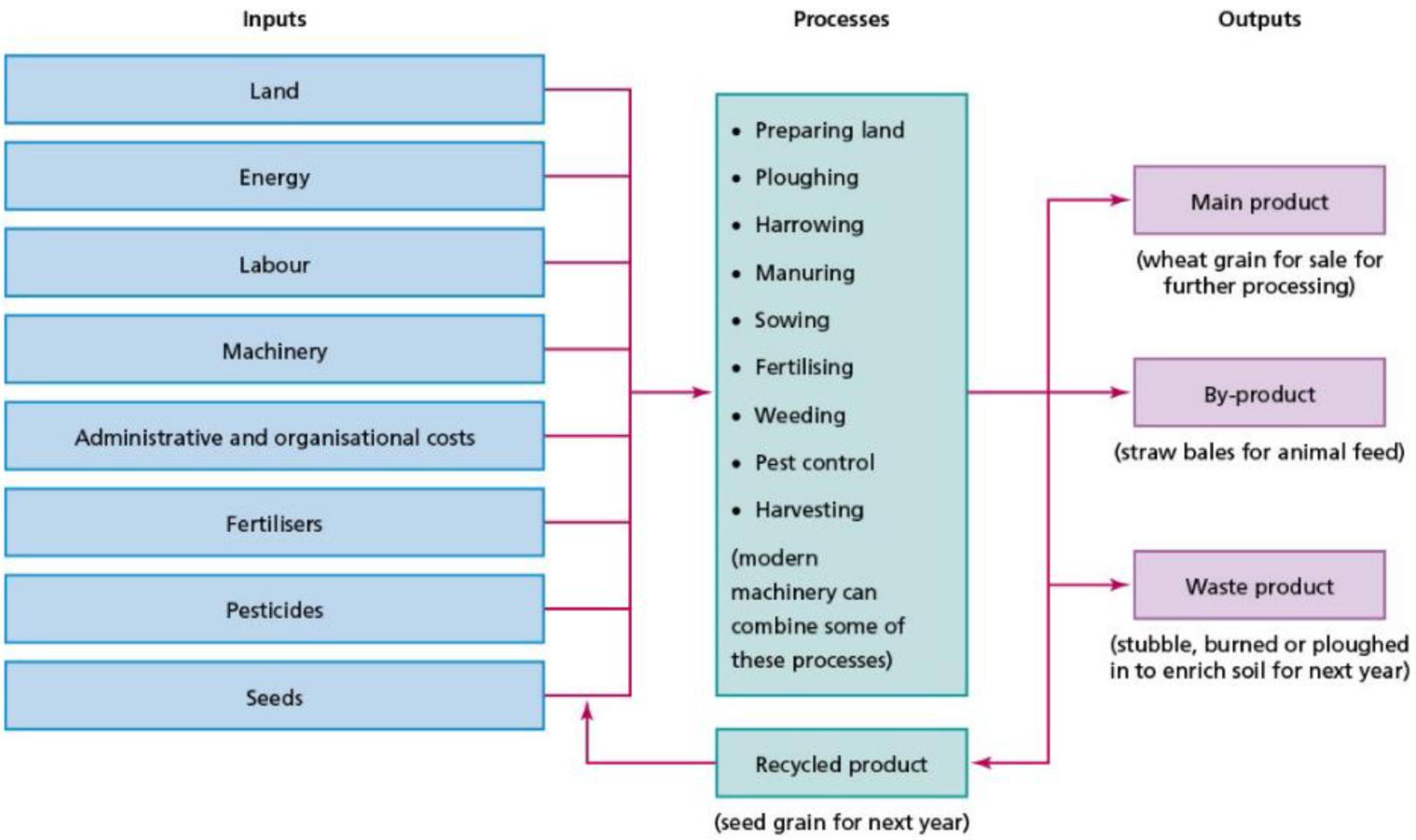
▲ **Figure 10.7** Vertical farming

Farming systems

Individual farms and general types of farming can be seen to operate as a **system**. A farm requires a range of **inputs**, such as labour and energy, to enable the **processes** that take place on the farm, such as ploughing and harvesting, to be carried out. The aim is to produce the best possible **outputs**, such as milk, eggs, meat and crops. A profit will be made only if the income from selling the outputs is greater than expenditure on the inputs and processes. Figure 10.8 shows an input–process–output (IPO) diagram for a wheat farm. Different types of agricultural system can be found within individual countries and around the world. Agricultural systems are dynamic human systems that change as farmers attempt to react to a range of physical and human factors.

Activities

1 **a** Explain the difference between arable farming and pastoral farming.
  b What is mixed farming?
2 Discuss the differences between:
  a commercial farming and subsistence farming
  b intensive farming and extensive farming.
3 Describe the characteristics of organic farming.
4 Describe the inputs, processes and outputs for a wheat farm (Figure 10.8).



▲ **Figure 10.8** The operation of a wheat farm